Cognitive Workload Allocation Under Uncertainty

A Stochastic Multiple Knapsack Approach with Workload-Dependent Returns

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Agenda

Outline

Cognitive Overload, Burnout

- Cognitive overload from excessive demands leads to billions of lost working days annually [Robinson, 2024].
- Productivity depends on attention management rather than rigid time scheduling [Grant, 2019].
- Traditional time management fails because it ignores the finite nature of human lifespan and cognitive capacity [Hart, 2024].
- Chronic illness, burnout risk, and work-family conflict underscore the need for systematic resource allocation that accounts for well-being [Cook and Zill, 2024, Yoo and Lee, 2022].

Motivation

- Cognitive overload and limited attention make time allocation fragile [Robinson, 2024, Grant, 2019].
- Sustainable planning needs explicit tradeoffs under finite time [Hart, 2024].
- Team settings add coordination and capacity coupling across agents.

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Problem Setting

Within a finite planning horizon, a team assigns tasks across multiple agents.

- Each agent has limited time, cognitive capacity, and attention (C_j).
- Task i assigned to agent j consumes stochastic size S_{ij} (realized effort).
- The realized return R_{ij} (utility) is also stochastic; it may depend on S_{ij} through

$$R_{ij} = g_{ij}(S_{ij}) + \varepsilon_{ij}, \quad \mathbb{E}[\varepsilon_{ij}] = 0.$$

- Deeper engagement can increase value, whereas overload may reduce returns.

Sets, Variables, and Parameters

Sets and indices

- Tasks $i \in \mathcal{I} := \{1, \ldots, N\}$
- Agents $j \in \mathcal{J} := \{1, \ldots, M\}$

Decision variables

$$x_{ij} = \begin{cases} 1, & \text{task } i \to \text{agent } j \\ 0, & \text{otherwise} \end{cases}$$

Stochastic characteristics

- S_{ij}: stochastic size (effort/attention)
- R_{ij}: stochastic return (utility/value)

Capacity and risk

- $C_j > 0$: agent capacity
- $\alpha \in (0, 1)$: risk tolerance

Stochastic Multiple Knapsack Formulation

$$
\begin{aligned}
\max_{x} \quad & \sum_{j \in \mathcal{J}} \sum_{i \in \mathcal{I}} \mathbb{E}[R_{ij}]\, x_{ij} & & \text{(Obj)} \\
\text{s.t.} \quad & \sum_{j \in \mathcal{J}} x_{ij} \le 1, & \forall i \in \mathcal{I} & \quad \text{(Assign)} \\
& \mathbb{P}\left(\sum_{i \in \mathcal{I}} S_{ij}\, x_{ij} \le C_j \right) \ge 1 - \alpha, & \forall j \in \mathcal{J} & \quad \text{(Chance)} \\
& x_{ij} \in \{0, 1\}, & \forall i, j. & \quad \text{(Binary)}
\end{aligned}
$$

(Chance) operationalizes cognitive/attention limits via chance constraints. Dependence between S_{ij} and R_{ij} captures workload-dependent productivity.

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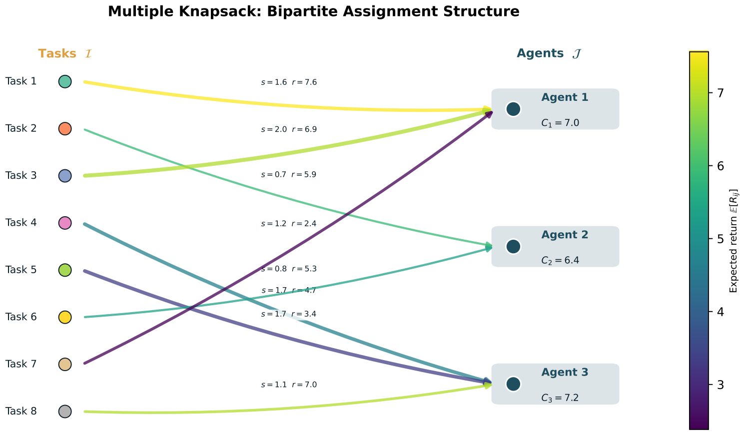
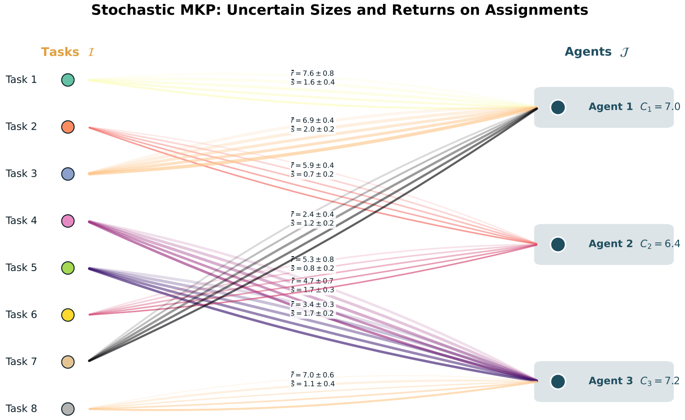
Multiple Knapsack Structure



Bipartite assignment: tasks (left) are assigned to agents (right) with capacities C_j. Edge width encodes task size $\mathbb{E}[S_{ij}]$ and color indicates expected return $\mathbb{E}[R_{ij}]$. Each task is assigned to at most one agent (constraint Assign).

Stochastic Sizes and Returns



Stochastic extensions: each assignment edge shows multiple scenario realizations of size S_{ij} (line width) and return R_{ij} (annotation). The chance constraint (Chance) limits the probability that total realized load exceeds agent capacity.

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Stochastic Knapsack: Cutting Planes and Dominance

- Büyüktahtakın [2023] derive scenario dominance cuts for multi-stage stochastic MIPs applied to the dynamic multi-dimensional knapsack. Strong dominance cuts reduce solution time with 0.06% average deviation from optimality.
- Büyüktahtakın [2022] generalize this via Stage-t Scenario Dominance for risk-averse (mean-CVaR) knapsack problems, achieving one-to-two orders of magnitude speedup.
- Joung et al. [2023] compare LP relaxation bounds for the robust knapsack problem and show that extended formulations yield identical bounds.

Distributionally Robust and Risk-averse Knapsack

- Bos et al. [2024] approximate the stochastic knapsack with periodic scheduled arrivals using Wasserstein DRO, producing robust healthcare surgical schedules.
- Ding et al. [2022] propose a target-based DRO knapsack with uncertain profit and capacity, reformulated as a second-order conic program.
- Merzifonluoglu and Geunes [2021] study the risk-averse static stochastic knapsack with CVaR and mean-standard deviation objectives, deriving efficient heuristics for normally distributed resource consumption.
- Boonstra et al. [2025] solve a robust multi-item newsvendor with budget constraint using a greedy knapsack algorithm under moment ambiguity.

Solution Algorithms for Stochastic Programs

- Chen and Luedtke [2022] analyze sample average approximation for two-stage programs without complete recourse, proving exponential convergence of recourse likelihood.
- Larsen et al. [2024] accelerate L-shaped methods via supervised ML, solving stochastic multi-knapsack instances in 20% of the time with less than 0.1% optimality gap.
- Lefebvre et al. [2024] reformulate adjustable robust optimization with discrete uncertainty and apply branch-and-cut to the multiple knapsack problem with uncertain weights.
- Lyu et al. [2022] unify responsive, adaptive, and anticipative allocation policies for the stochastic knapsack through persistency values.

Assignment, Scheduling, and Workforce Planning

- Rezaeian et al. [2024] model the stochastic assignment of projects to managers via multi-stage stochastic integer programming with heuristic algorithms for large instances.
- Nasirian et al. [2025] solve multiskilled workforce staffing and scheduling via logic-based Benders decomposition, achieving 29x speedup over direct MILP solving.
- Cavagnini et al. [2020] use two-stage stochastic programming for production planning under uncertain learning rates, with insights on cross-training and worker assignment.
- Novak et al. [2022] connect stochastic parallel machine scheduling to an inflatable stochastic knapsack pricing subproblem within branch-and-price.

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Possible solution methodology

- Benders decomposition:
  - Master problem: choose task selection and assignment decisions $x_{ij} \in \{0, 1\}$.
  - Subproblems: for each agent/scenario, compute overload/risk and generate Benders cuts.

Possible expected results

- R1: Return-Risk trade-off.
  - As risk tolerance α decreases, optimal expected return decreases, while overload violation rate drops sharply.
- R2: Allocation pattern changes.
  - Tighter α makes the system more distributed: tasks are spread across agents to avoid single-point overload.
  - Fewer tasks with high variance / highly uncertain durations are selected.
- R3: Impact of stress-performance dependence.
  - If $R_{ij} = g_{ij}(S_{ij}) + \varepsilon$ and g_{ij} decreases under high workload, the optimal policy avoids near-capacity assignments more aggressively.

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Key Takeaways

- The formulation models cognitive resource allocation as a multiple knapsack problem with stochastic sizes and returns, capturing workload-dependent productivity via $R_{ij} = g_{ij}(S_{ij}) + \varepsilon_{ij}$.
- Chance constraints operationalize cognitive/attention limits, controlling overload risk while retaining assignment flexibility.
- Scenario dominance cuts [Büyüktahtakın, 2023], DRO reformulations [Bos et al., 2024, Ding et al., 2022], and ML-accelerated decomposition [Larsen et al., 2024] are natural tools for scaling the model to organizational settings.
- Data-driven SAA [Chen and Luedtke, 2022] and risk-averse objectives [Merzifonluoglu and Geunes, 2021] provide robust solution strategies under limited distributional information.

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